

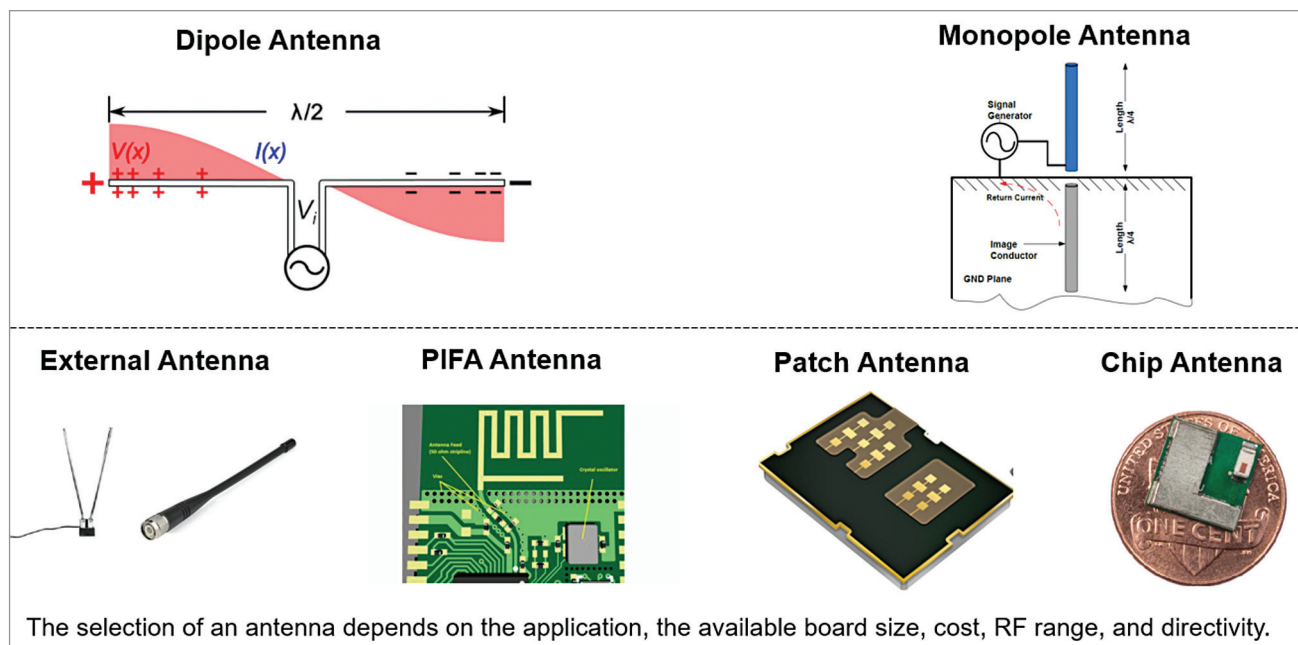


Fig. 4: Types of antennas

To explain it simply: Hosted designs split tasks between the wireless chip and the processor, offering control but adding complexity. Hostless designs put everything in one chip, offering simplicity and faster deployment.

Choosing between the two depends on your priorities. If you want faster time-to-market and fewer development headaches, hostless is often the way to go. If you are building something more advanced and need tighter control over how the wireless system behaves, hosted offers more freedom, but at the cost of added complexity.

The signal superhero: Antenna types

Antennas are a critical part of RF communications, and this is where many new engineers go wrong. While dipole and monopole antennas are often cited in academia, real-world wireless products rely on more evolved designs. The choice depends largely on performance requirements and physical constraints.

External antennas. Designed to be protruding outside, these offer the highest efficiency, gain, and directivity. However, they take up

space, making them less suitable for compact devices.

PCB trace antennas. Cheap and etched directly onto the board, these are widely used in multi-band designs but are less efficient than external antennas. So, the efficiency of their offering is moderate compared to external antennas, and they are widely adopted in multi-band designs. Also, these are among the cheapest antennas available.

Patch antennas, complex but high-performing, are used in MIMO and radar systems. Despite their performance advantages, their complexity and design overhead make them better suited for specialist use cases.

Chip antennas, the smallest and used commonly in mobile and tracking devices, are highly integrated but have reduced gain and range. Their miniaturisation is achieved using high-dielectric materials, enabling sizes as small as one-tenth that of a comparable PCB antenna.

Hardware hacks: To avoid fry and cry

Beyond antenna selection, RF product design hinges on precision,

especially in high-frequency systems. One of the most critical aspects is impedance matching, which ensures maximum power transfer between RF components, such as ICs, filters, and antennas.

Using a simple analogy: just as water flows best through a pipe with a constant diameter, electrical signals flow optimally through circuits with consistent impedance. Disruptions cause reflections and wasted energy. In practice, designers aim to match the output impedance of the source and the input impedance of the load to a common characteristic impedance, typically 50 ohms in most RF systems.

Successful RF design also involves careful attention to filtering, grounding, component selection, and PCB layout. All of these influence signal integrity and overall system performance. While the concepts may seem intricate, mastering them is crucial for developing robust and efficient wireless products.

Power supply matching

At high frequencies, conventional design assumptions no longer hold. A standard wire, perfectly functional in DC circuits, begins to behave like a